

Samarth Saxena

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EXPERIENCE

Siemens Technology & Services [\[Certificate\]](#)

July 2024 – Feb 2025

Software Engineer Intern, Full Stack & AI

Hybrid

- Built SkillX, an enterprise assessment platform serving 500+ daily users, and designed a 3-tier role-based access control system in Node.js, Express, and MySQL that cut batch management overhead by 40%.
- Engineered a real-time AI proctoring system with TensorFlow.js and BlazeFace that flagged head movement and fullscreen exits at under 200 ms alert latency.
- Hardened the backend with JWT authentication and tuned connection pooling, holding 99.9% uptime through peak exam load.

Centre for Neuroinformatics (CNI), SRIP 2026

May 2026 – July 2026

Research Intern, NVIDIA HPC & Neuro-QML Track

Remote

- Selected from a national applicant pool for a research program pairing multimodal brain-signal analysis (EEG, NIRS) with generative AI.
- Trained hybrid quantum-classical models on terabyte-scale neurological datasets across an 8x NVIDIA A100 GPU cluster, targeting low-latency inference for clinical prediction.
- Built an agentic Neuro-QML platform on NVIDIA BioNeMo biomedical LLMs with automated validation agents to raise the explainability and accuracy of clinical outputs.

PROJECTS

Citadel, Zero-Trust LLM Security Middleware [\[GitHub\]](#) | FastAPI, Next.js 14, Lark, Docker

- Built a high-throughput security proxy that screens every prompt for injection and jailbreak attempts, signs each payload with HMAC-SHA256, and structurally validates the model's output before it reaches the app, adding under 200 ms of round-trip overhead.
- Designed a Lark LALR(1) context-free grammar parser inside a two-stage, fail-closed validation pipeline that rejects malformed or unsafe traffic outright instead of repairing or guessing.
- Benchmarked with a 500-request async suite and 216 pytest cases at a 100% pass rate in 1.37 s, then shipped it as a multi-stage Docker image running non-root with a read-only filesystem and all Linux capabilities dropped.

LLM From Scratch, 71.5M-Parameter GPT [\[GitHub\]](#) | PyTorch, Flask, tiktoken

- Pretrained a 71.5M-parameter decoder-only GPT on ~0.7B FineWeb-Edu tokens across 2x NVIDIA A100 80GB GPUs in 84 minutes, reaching a 3.49 validation loss.
- Implemented a modern transformer stack from scratch (RoPE embeddings, RMSNorm, ReLU² MLP, QK-norm, tied embeddings, logit soft-cap), trained with the Muon optimizer on hidden layers and AdamW on embeddings.
- Instruction-tuned the base model on a Dolly-15k SFT mix and served it through a Flask chat app with an optional retrieval mode that grounds answers in fetched web context.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C++, Java, SQL (MySQL, PostgreSQL)

AI & ML: LLM Pretraining & Fine-Tuning, Transformers, RAG, Multi-Agent Systems, PyTorch, TensorFlow.js, LangChain, FAISS, scikit-learn

Backend & Security: FastAPI, Node.js, Express, ASGI Middleware, HMAC-SHA256, JWT, Lark CFG, Static Analysis, Zero-Trust Architecture

Infrastructure: Docker, Docker Compose, NVIDIA Triton, AWS, Git/GitHub

Frontend: React.js, Next.js 14, Tailwind CSS, shadcn/ui

Tools: VS Code, Postman, Pytest, Uvicorn, Vercel, Railway

ACHIEVEMENTS & CERTIFICATIONS

- **1st Place, ThinkBridge Hackathon:** led a team of 4 to build an AI solution, placing 1st out of 200+ competing teams.
- **Oracle Certified Generative AI Professional (2025):** LLM fundamentals, prompt engineering, and production generative AI development.
- **AWS Solutions Architect, Associate (Expected June 2026):** preparing for the SAA-C03 exam.

EDUCATION

Vellore Institute of Technology, Chennai

Sept 2023 – May 2027

B.Tech in Computer Science (AI & Robotics), CGPA: 8.41 / 10.0

Aditya English Medium School, Pune, HSC (Grade 12): 78.2%

2022 – 2023

Vibgyor High School, Pune, SSC (Grade 10): 94.1%

2020 – 2021